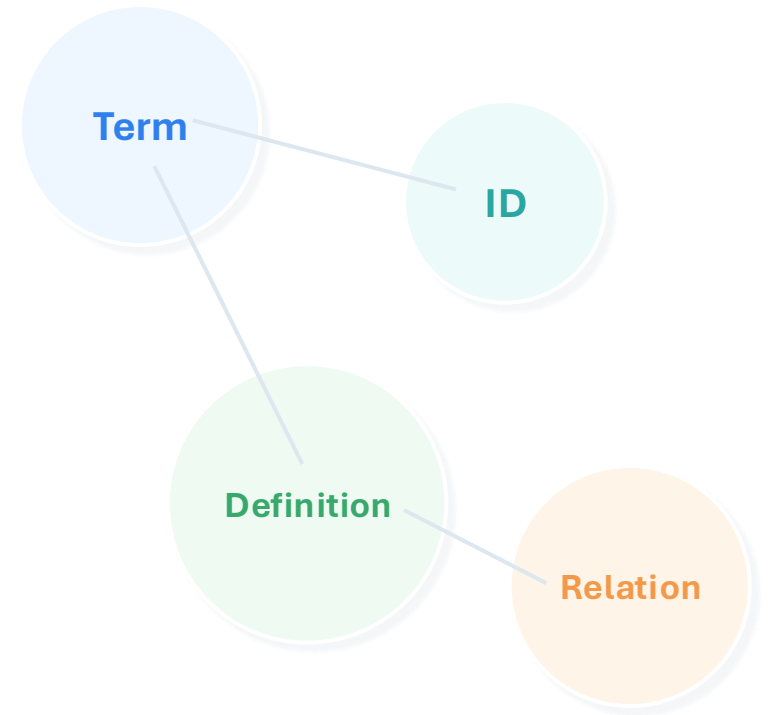


Ontology Concepts for Chado

From biological meaning to cvterm

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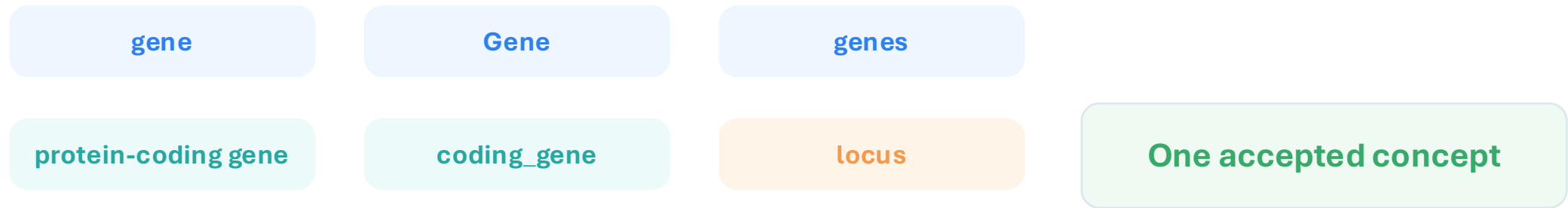


cvterm

semantic bridge

The problem: biology has many names for one idea

Humans can infer that these labels are related. Databases usually cannot.



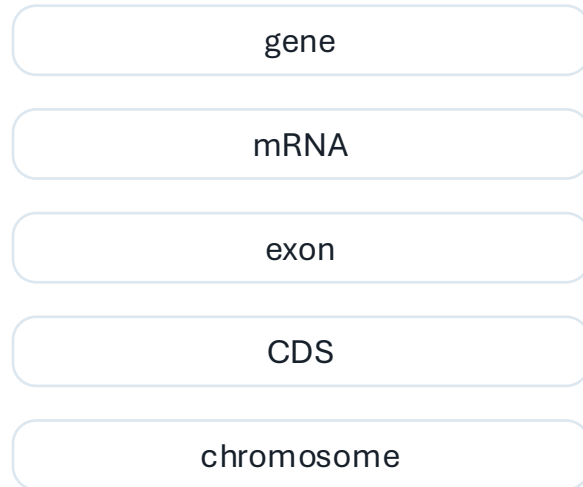
Controlled terms reduce ambiguity, typos, and local naming habits.

Chado uses this idea by linking biological records to controlled vocabulary terms instead of relying only on free text.

Controlled vocabulary vs ontology

A controlled vocabulary standardizes words. An ontology standardizes meaning.

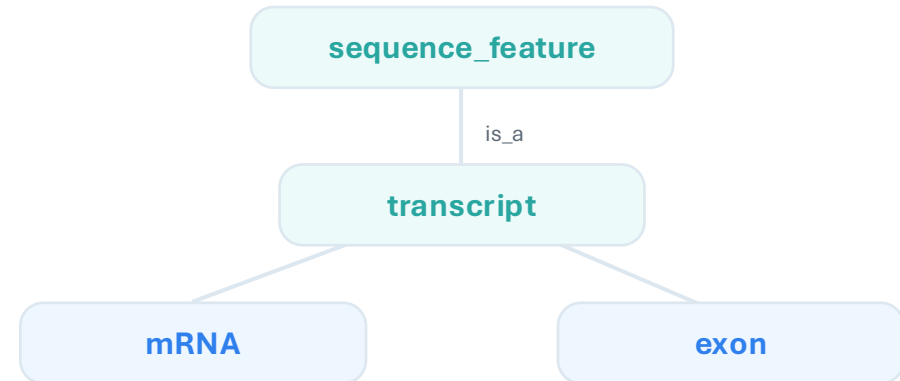
CONTROLLED VOCABULARY



A list of accepted terms

Useful for consistency and validation.

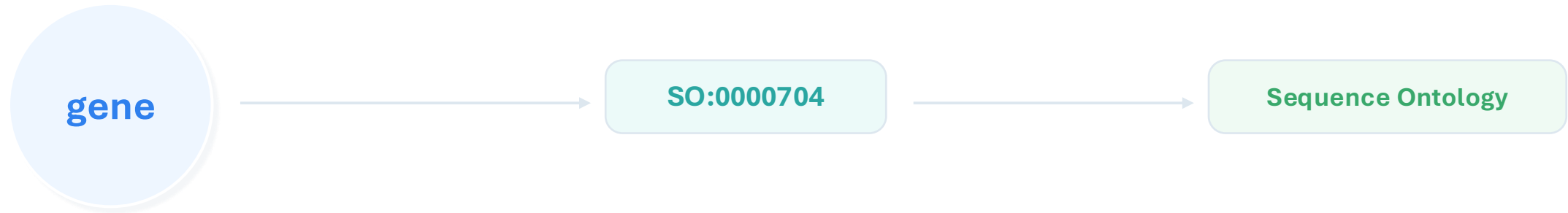
ONTOLOGY



Terms + definitions + relationships

Useful for meaning, integration, and broader queries.

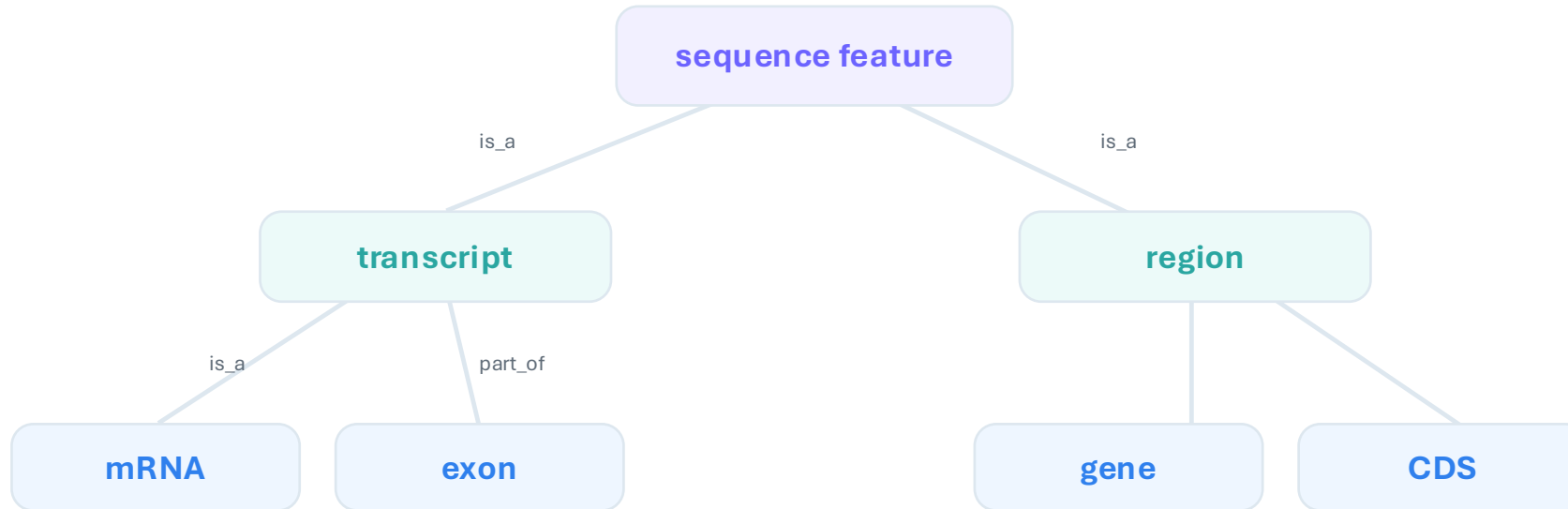
An ontology term is a defined concept, not just a word



Component	Example	Why it matters
Name	gene	Readable label
Stable ID	SO:0000704	Persistent reference
Definition	sequence region with function	Limits ambiguity
Relationships	is_a biological_region	Places concept in context
Synonyms	locus, coding gene	Helps map alternatives

Relationships turn terms into a biological graph

The power of an ontology comes from edges between concepts.



Reasoning example

A query for a broad parent term can include records annotated to more specific child terms.

Common ontology relationship types

Relationship types are also controlled terms. They describe how concepts connect.

is_a

Subtype of a broader concept

mRNA is_a transcript

part_of

Component of another concept

exon part_of transcript

derives_from

One entity comes from another

protein derives_from mRNA

regulates

One process affects another

regulation of expression

Chado stores relation labels as cvterms too, allowing the vocabulary of relationships to be extended.

Ontologies commonly used around Chado

Different biological questions often need different vocabularies.

so

[Sequence Ontology](#)

feature types: gene, mRNA, exon, CDS

GO

[Gene Ontology](#)

function, process, cellular component

RO

[Relation Ontology](#)

part_of, derives_from, regulates

TO /
PO

[Trait](#) / [plant](#) ontologies

traits, phenotypes, plant structures

Local
CV

Project-specific vocabulary

terms not covered by public ontologies

co

[Crop Ontology](#)

CGIAR community developed ontologies



Public ontology when possible. Local CV when needed.

Rice Ontology

DOI [10.5281/zenodo.8253490](https://doi.org/10.5281/zenodo.8253490)



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Collaborators



Alliance



Navigation

TE=Term, **TR**=Trait, **ME**=Method, and **SC**=Scale

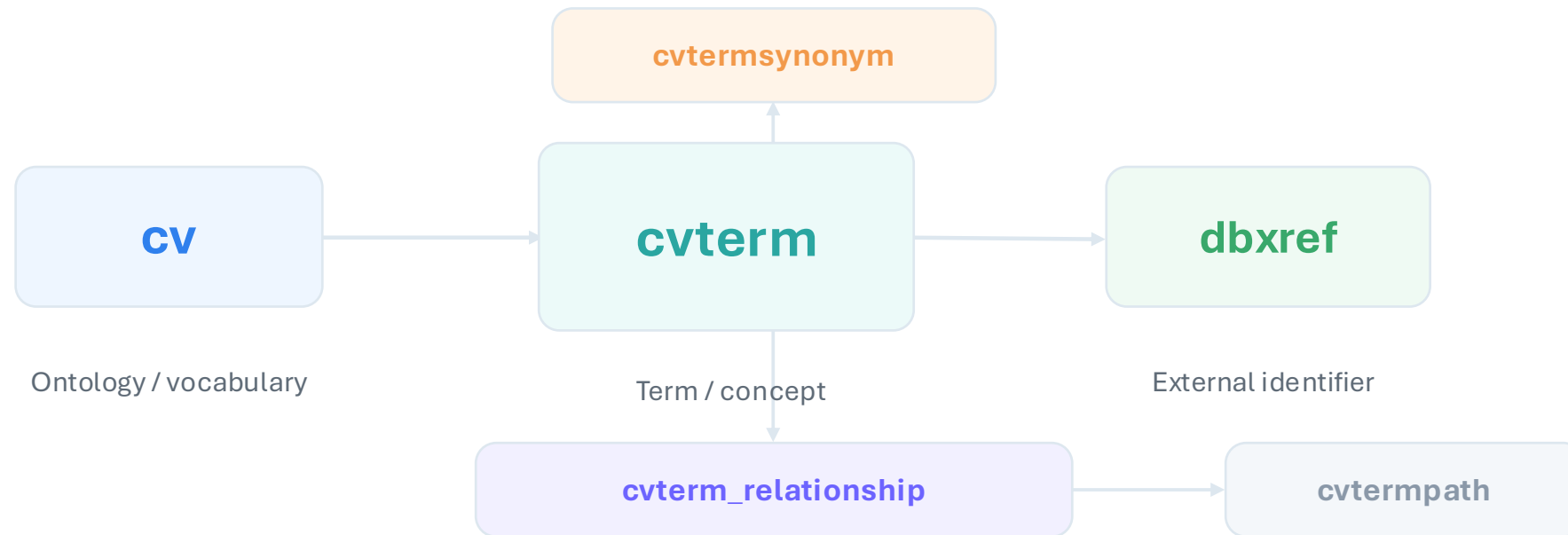
- TE Rice traits
 - TE Abiotic stress
 - TE Agronomic
 - TE Biochemical
 - TE Biotic stress
 - TE Fertility
 - TE Morphological
 - TE Phenological
 - TE Physiological
 - TE Quality

Concept details

Key	Value
term_id	CO_320:ROOT
ontology_id	CO_320
ontology_name	Rice
language	EN

How Chado represents ontology concepts

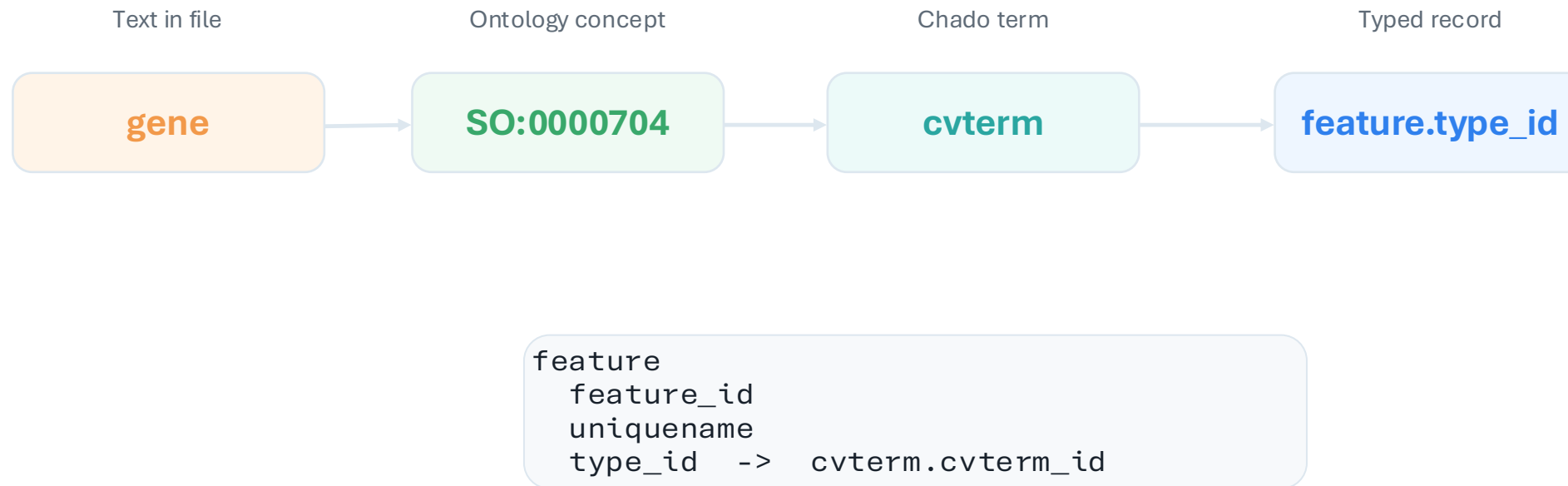
The CV module stores vocabularies, terms, identifiers, synonyms, and relationships.



In Chado, cvterm is the central object that many other tables point to.

cvterm is the semantic bridge

A biological record becomes meaningful when it points to the right term.

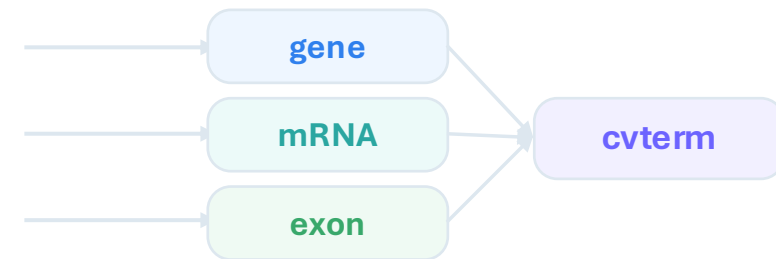


This is why typing a feature as a gene is not just a label; it is a link to a defined ontology concept.

Example: genome annotation terms

A GFF-like file contains strings. Chado needs those strings to resolve to terms.

```
Chr1  source  gene  1000  5000  .  +  .  ID=gene1
Chr1  source  mRNA  1000  5000  .  +  .  ID=tx1;Parent=gene1
Chr1  source  exon  1000  1200  .  +  .  Parent=tx1
```



Loader question

Can every feature type and relationship in the file be matched to the expected ontology term?

Why this matters for loading and querying

The CV module supports both data quality and biological interpretation.

1

Data integrity

Avoid misspellings, duplicates, and informal labels.

2

Portability

Use shared ontology terms so datasets can be integrated.

3

Reasoning

Use parent-child relationships to query broad concepts.

4

Troubleshooting

Trace loader errors to missing, wrong, or ambiguous terms.

Most cvterm-related failures are term-resolution problems.

Mini case: when a term does not resolve

A loader reports: Could not find cvterm "lncRNA"

1

What kind of concept is it?

Feature type, relationship type, or property?

2

Which ontology should contain it?

For sequence features, often Sequence Ontology.

3

Is the accepted name different?

Maybe lnc_RNA rather than lncRNA.

4

What should be fixed?

Input mapping, ontology loading, or local CV.

Key decision: correct the input, load/update the ontology, or add a local term.

Take-home messages

- 1 A controlled vocabulary standardizes allowed terms.
- 2 An ontology adds definitions, identifiers, and relationships.
- 3 Chado stores ontology concepts through the CV module.
- 4 **cvterm connects biological records to standardized meaning.**
- 5 Loader errors often reveal missing, wrong, or ambiguous term mappings.

Concept pathway biological idea -> ontology term -> cvterm -> typed Chado record

Source notes

Core sources used to align the presentation with Chado:

1

GMOD Chado Tutorial - CV (Controlled Vocabularies) Module

https://gmod.org/wiki/Chado_Tutorial#CV_.28Controlled_Vocabularies.29_Module

2

GMOD Chado CV Module

https://gmod.org/wiki/Chado_CV_Module

3

Chado schema documentation - cv module

https://chado.readthedocs.io/en/rtd/_static/schemaspy_integration/cv/

Suggested discussion prompt: What local terms in your project need formal CV treatment?